Stanford CS224V Course
Agentic AI

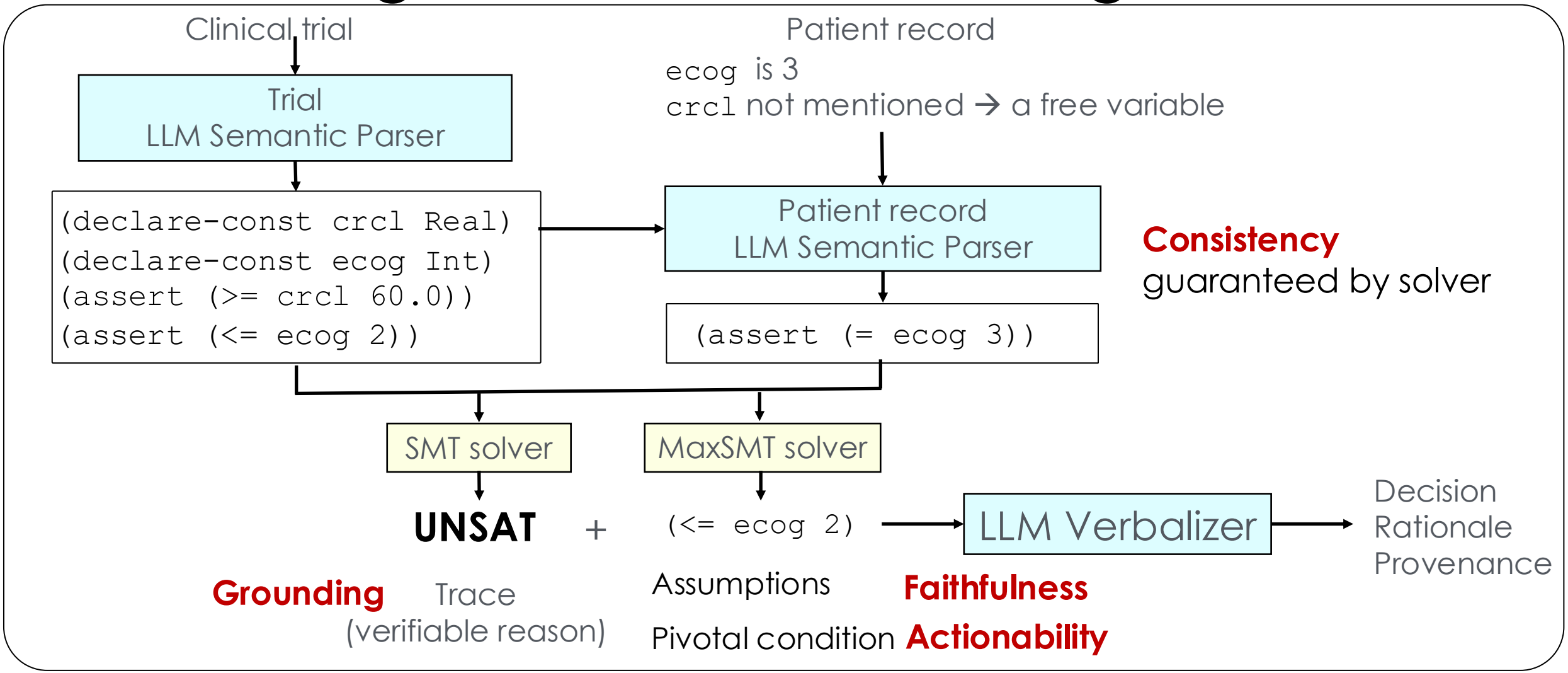
Lecture 4

Constraint-Satisfaction Based Retrieval

Case study: Clinical Trial Matching

Cyrus Zhou and Monica Lam

Formulating clinical trial matching



Cost of VERDICT

m: number of patients; n: number of trials	#times
• Trial parsed only once	O(n)
• Patient record is parsed differently for each trial	O(mn)
• SMT, MaxSMT calls	O(mn)

Too expensive for 500,000 clinical trials per patient!

Can we use a cheaper retrieval method to come up with a short list for matching?

Embedding-based retrieval has poor recall – may miss life-saving trials.

Goal: (1) Perfect recall

(2) High accuracy

New idea: SatIR, **Constraint-satisfaction based retrieval**

Key Idea in SatIR

m: number of patients; n: number of trials

	#times	SatIR
• Trial parsed only once	O(n)	O(n)
• Patient record is parsed differently for each trial	O(mn)	O(m)
• SMT, MaxSMT calls	O(mn)	2.5K trials: 0.14ms/patient

KEY: Patient records are coded once for all trials

Too expensive for 500,000 clinical trials per patient!

SatIR: Use relational algebra (Databases): **less expressive**, but **much cheaper**

Project SMT formulas to relational algebra

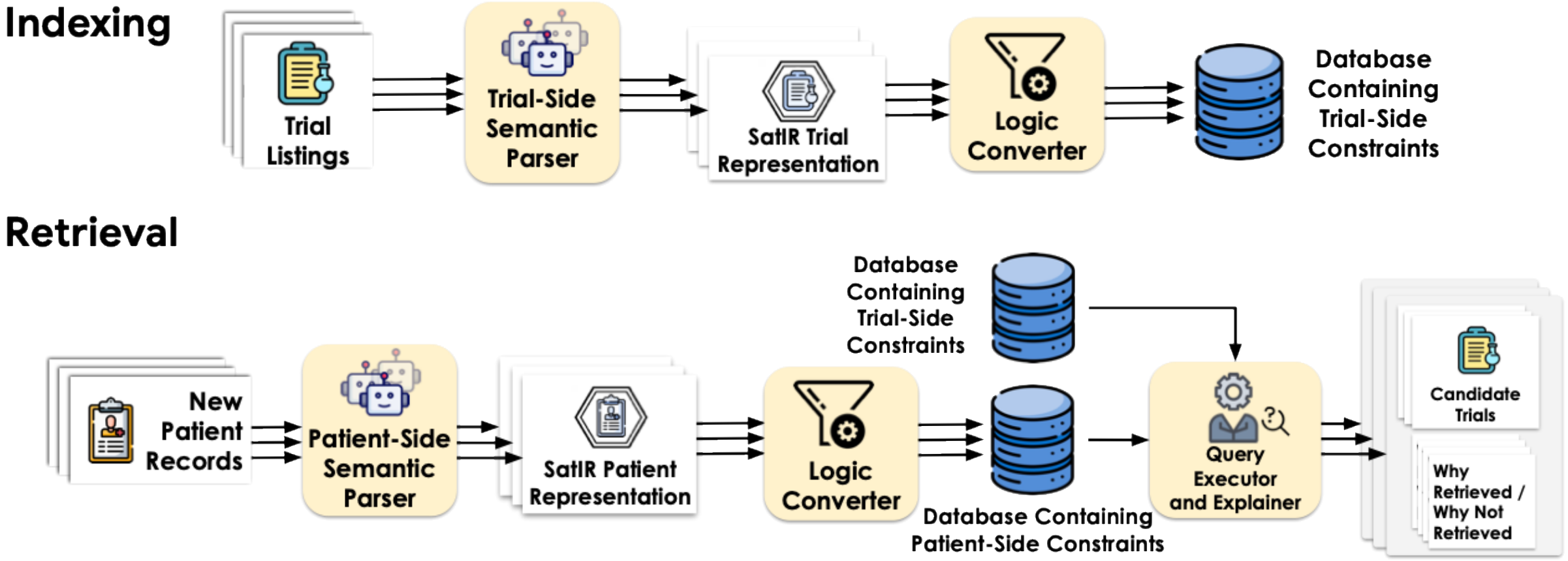
An important math concept



PERFECT RECALL $S \subseteq D$

INACCURACY $D - S$

SatIR: Satisfiability-based information retrieval



The constraints are stored in the databases

Why is this difficult?

1. SMTs symbols have no meaning!
   - Every nuance and inference needs to be spelled out

2. How to turn SMT satisfaction into database operations?

Outline of this Lecture

1. High-level concept in SatIR
2. Semantic parsing: NL to SMT for clinical trials
   - Basic concepts
   - Advanced concepts
3. Fast constraint-satisfaction based retrieval
4. Experimental Results

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What is a Semantic Parser?

Translates NL into some formal representation



Formal Semantics: SMT Representation

An SMT formula: a set of atoms in a propositional formula (and, or, not)

For clinical trial matching:

```
atom = (variable, cmp, target)
variable = (relation, concept, qualifier)
((hasCondition, Type2Diabetes, within5y), =, true)
((hasMeasurement, HbA1c, current), <, 7.0)
```

Examples of Qualifiers

```
"steroid-induced osteonecrosis of the jaw" ⟶
    ✗ steroid_induced, osteonecrosis, jaw
    ✓ base: Osteonecrosis
      @@ qualifier: steroid-induced
      @@ qualifier: finding site = jaw
```

```
patient: type2_diabetes diagnosed three years ago
fact:    type2_diabetes @@ [t-3y, now]

trial:       type2_diabetes @@within5y
constraint: interval ∩ [now-5y, now] ≠ ∅
```

A parsed trial (Kawasaki disease, SatIR Fig. 3)

Kawasaki syndrome

with coronary artery dilation or aneurysm,

and ≥ 1 of:

coronary Z-score ≥ 2.5 in any major segment;

Persistent or recrudescent fever
≥ 38.0 °C
for ≥ 36 h after initial therapy;

or evidence of active inflammation.

```
(assert (and kawasaki_now
  (or coronary_dilatation_now
      coronary_aneurysm_now)))
(assert (or
  (and z_score_recorded_now@@any_major_segment
       (>= z_score 2.5))
  (and (or fever_now@@persistent
           fever_now@@recrudescent)
       (>= body_temp_celsius 38.0)
       fever_now@@36h_after_initial_therapy)
  acute_inflammation_now))
```

SMT programs are complex

Each trial is represented by a non-trivial SMT program.

Typical number of lines:

~360

One-shot text-to-SMT is brittle

Problems

- Variables redeclared inconsistently
- Qualifiers dropped or misattached
- Polarity flips (esp. exclusions)
- Valid syntax, wrong meaning

Techniques

- Split cohorts
- Reverse exclusion polarity → one list
- Make entity spans contiguous
- Make and/or precedence explicit
- Split into self-contained units

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Disadvantages of SMT Approach

- SMT solvers operate on variables – they have no meaning
- All information useful for SMT solvers must be included formally

1. **Taxonomy**: same concepts → same variable name **Canonicalization**

2. **Implication relations** between concepts
   e.g. A ⇒ B means A satisfies B
   Include implication relations

3. **Missing Data:** most conditions in trials are missing **Data Imputation**

Challenge 1: Canonicalization

Theorem provers operate on variables – they have no meaning

- Same concepts must be represented by the same variables

Example

	Patient p	**Trial** t	$p \wedge t$
English	Has dizzy spells	Has syncope	SAT
Semantic parse	Has-dizzy-spells	Has-syncope	UNSAT
Canonicalized	Has-syncope	Has-syncope	SAT

Canonicalization is complex! So many clinical terms!

Standardization in medicine are important → SNOMED

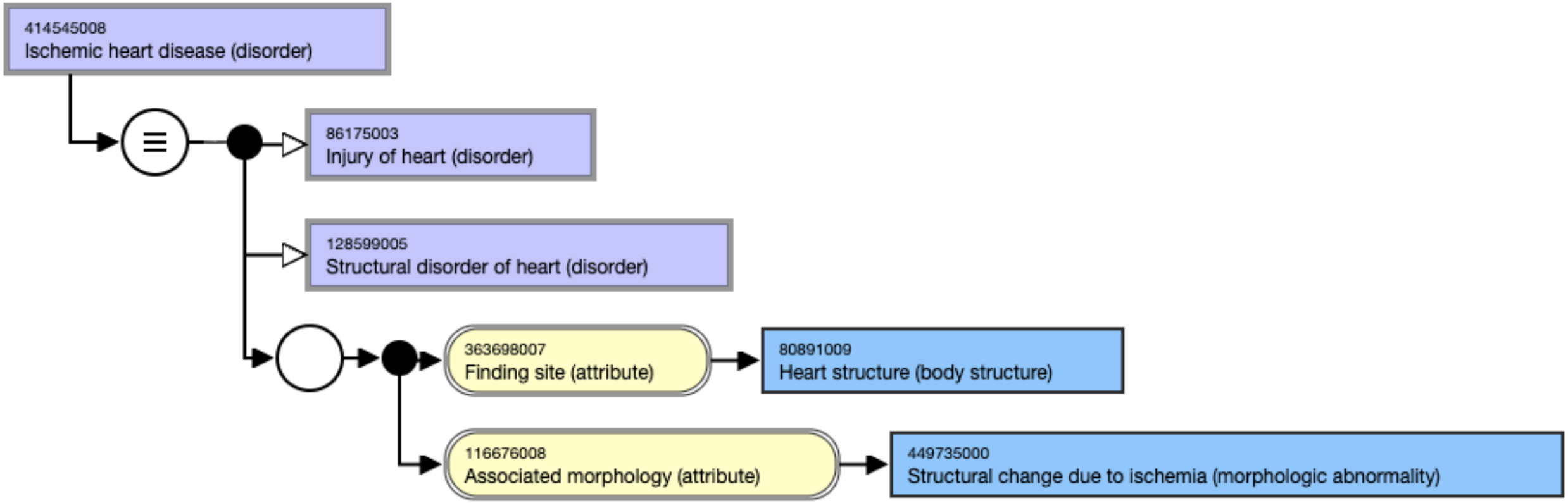
SNOMED
Systematized Nomenclature of Medicine – Clinical Terms

US Edition of SNOMED

- 300,000+ unique concepts
- Over 1,000,000 descriptions
- 903,000 links or semantic relationships between concepts.

SNOMED Example

- Top-level classes: Clinical Findings, Procedures, Observable Entities, Substances…



Using SNOMED in the SMT representation

An SMT formula: a set of atoms in a propositional formula (and, or, not)

For clinical trial matching:

```
atom = (variable, cmp, target)
variable = (relation, concept, qualifier)
```

Relations: (we defined them empirically)

- e.g. hasFindingOf, hasCondition, hasMeasurement, hasUndergone

Canonical concepts

- Use canonical terms from SNOMED in clinical trials and patient records

Non-canonical concepts

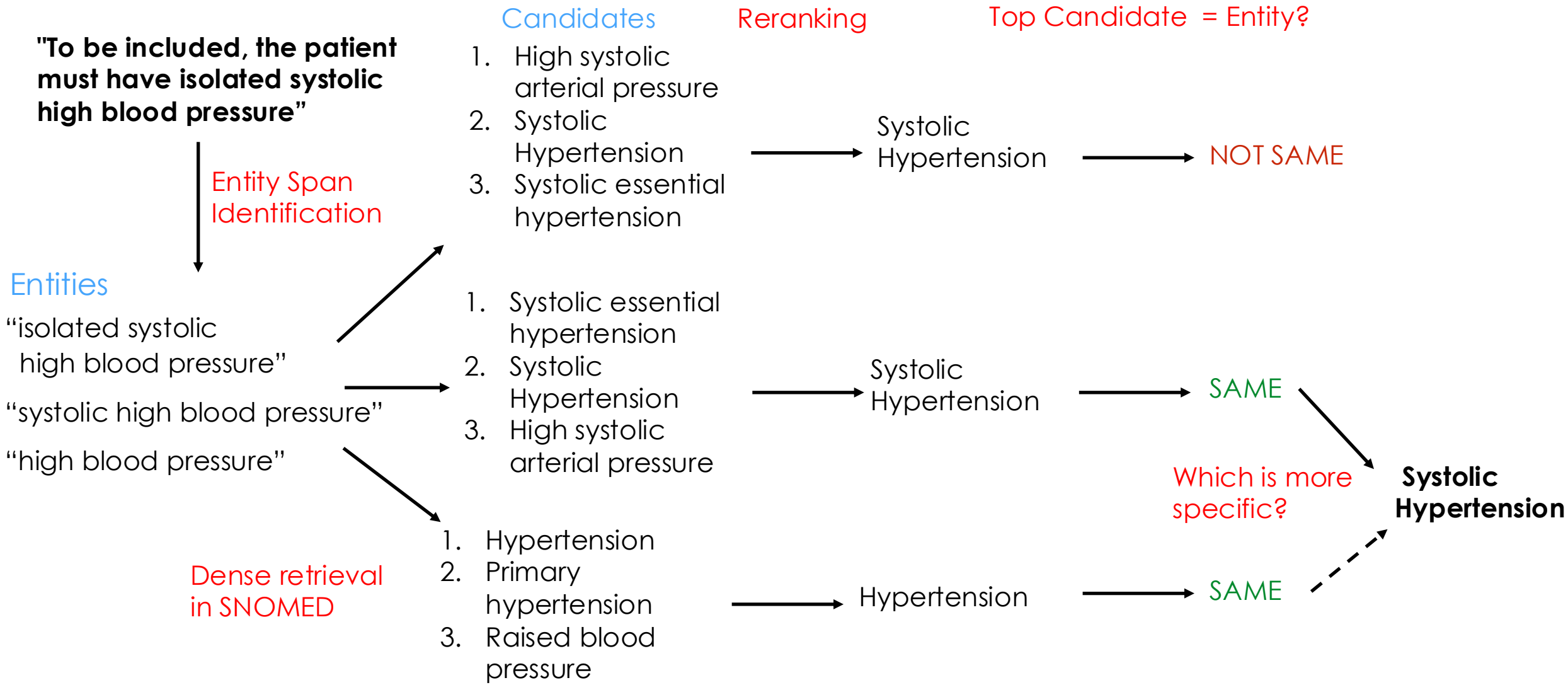
- Many concepts are trials/patient records are NOT in SNOMED
- Make up non-canonical variables with a description
- E.g. HasAdequateSocialSupportForFollow

Canonicalization Algorithm

- Problem: NER (Named entity recognizer)
  - Turn the informal English to the closest term in SNOMED

- Challenges and solutions:
  - Many similar terms in SNOMED → Dense retrieval
  - Find the most **precise** entity in SNOMED →
    Retrieve, rerank, ensure equivalence, pick the most precise

How to Canonicalize?



A Full Trial Example

The patient must have Kawasaki syndrome with either coronary artery dilation or aneurysm and meet one of the following conditions:
- coronary artery involvement, defined as Z-score ≥ 2.5 in any major coronary segment;
- persistent or recrudescent fever, defined as temperature ≥ 38.0°C for ≥ 36 hours after initial therapy; or
- evidence of active inflammation,

i. Preprocessing of Free Text

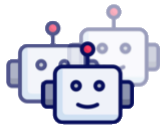


To be included, the patient must have Kawasaki syndrome with **(**coronary artery dilation**) OR (coronary artery** aneurysm**)**.

To be included, the patient must satisfy **≥ 1** of the following conditions:
- **(**coronary artery involvement, defined as Z-score ≥ 2.5 in any major coronary segment**) OR**
- **(**persistent fever **OR** recrudescent fever, defined as temperature ≥ 38.0°C for ≥ 36 hours after initial therapy**) OR**
- **(**evidence of active inflammation**)**

SNOMED-Based Medical Ontology

ii. Canonicalization of Medical Concepts



Dilation:
- Dilatation of Blood Vessel
- ID: 248493003

Coronary Artery Aneurysm:
- Aneurysm of Coronary Vessels
- ID: 50570003

Coronary Artery:
- Coronary Artery
- ID: 41801008I

Fever:
- Fever
- ID: 386661006

Body Temperature:
- Body Temperature
- ID: 386725007

Active inflammation:
- Acute Inflammation
- ID: 4532008

iii. Programming of Constraints



```
(assert (and
acute_febrile_mucocutaneous_lymph_node_syndrome_now
        (or
dilatation_of_blood_vessel_now@@finding_site_coronary_artery
            aneurysm_of_coronary_vessels_now)))
(assert (or (and
coronary_artery_z_score_value_recorded_now_with_unit_unitless@@in_any_major_coronary_segment
            (>=
coronary_artery_z_score_value_recorded_now_with_unit_unitless 2.5))
        (and (or fever_now@@persistent
fever_now@@recrudescent)
            (>=
body_temperature_value_recorded_now_withunit_celsius 38.0)
            fever_now@@foradurationof36hours

fever_now@@temporalcontext_after_initial_therapy)
        acute_inflammation_now))
```

Text Color Legend

Preprocessing Modifications	Canonicalized Concepts	Canonical SMT Variables	Qualifiers to Canonical Variables	Non-Canonical Variables

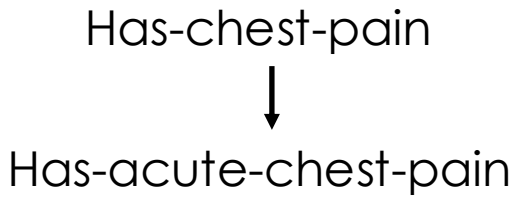
Challenge 2: Implication Relations Between Concepts

Example

	Patient p	**Trial** t	$p \wedge t$
Canonicalized	Has-acute-chest-pain	Has-chest-pain	UNSAT
With implications	Has-acute-chest-pain (Has-chest-pain)	Has-chest-pain	SAT

- **SNOMED includes implication relations between terms:**

  ∃ an edge (x,y) if y ⟹ x



Has-acute-chest-pain ⟹ Has-chest-pain

¬ Has-chest-pain ⟹ ¬ Has-acute-chest-pain

- Codifying the patient
  - If patient has x, includes ancestors of x
  - If patient has ¬ x, includes the ¬ of all descendants of x

Different kinds of subsumption

- Concept: AcuteAppendicitis ⊑ Appendicitis
- Relation: HasDiagnosisOf ⊑ HasFindingOf
- Qualifier: specific ⊑ general
- Causal: postop hemorrhage ⊑ underwent surgery

The patient records include all the implications

An important concept

–42%

True positives
if we don't use the ontology

(TREC 2022, GPT-5 judge)

Challenge 3. Missing Data

Use **salience** of property in question to impute data
Assume: Important facts get written down

Trial asks about	If missing, assume:	Why
Pregnancy	false	Pregnancy is salient
Ability to walk	true	Inability to walk is salient
Hospitalized now	unconstrained	Could go either way

Per-condition LLM question

"If this were true,
would it plausibly be in the note?"

1,190

stored default values (TREC)

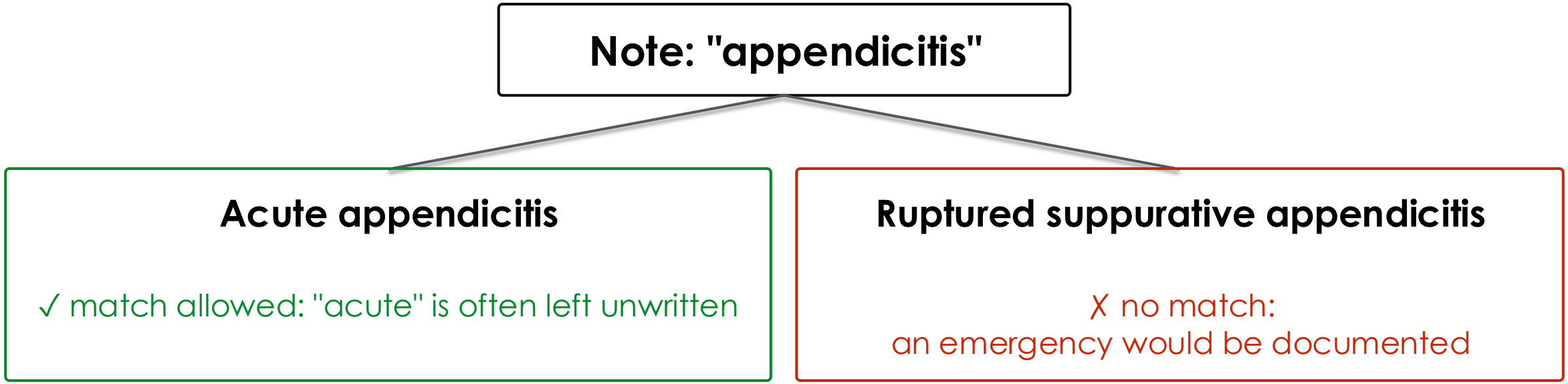
–7 to –9%

true positives without them

Using salience in handling specificity:

when may a vague note match a specific trial?

Hierarchy of diseases



Widen to a more specific trial target only if the detail could plausibly be omitted.

Other constraints

Trial constraints need to be classified as hard, soft, operational (ignored)

Explicit matching objective

Objective	**A trial matches if it…**
treat-chief	treats the chief complaint
treat-any	treats any documented condition
relevant-to-any	is meaningfully related to any condition

The objective is a formula too: a product decision, made explicit.

Size of patient records

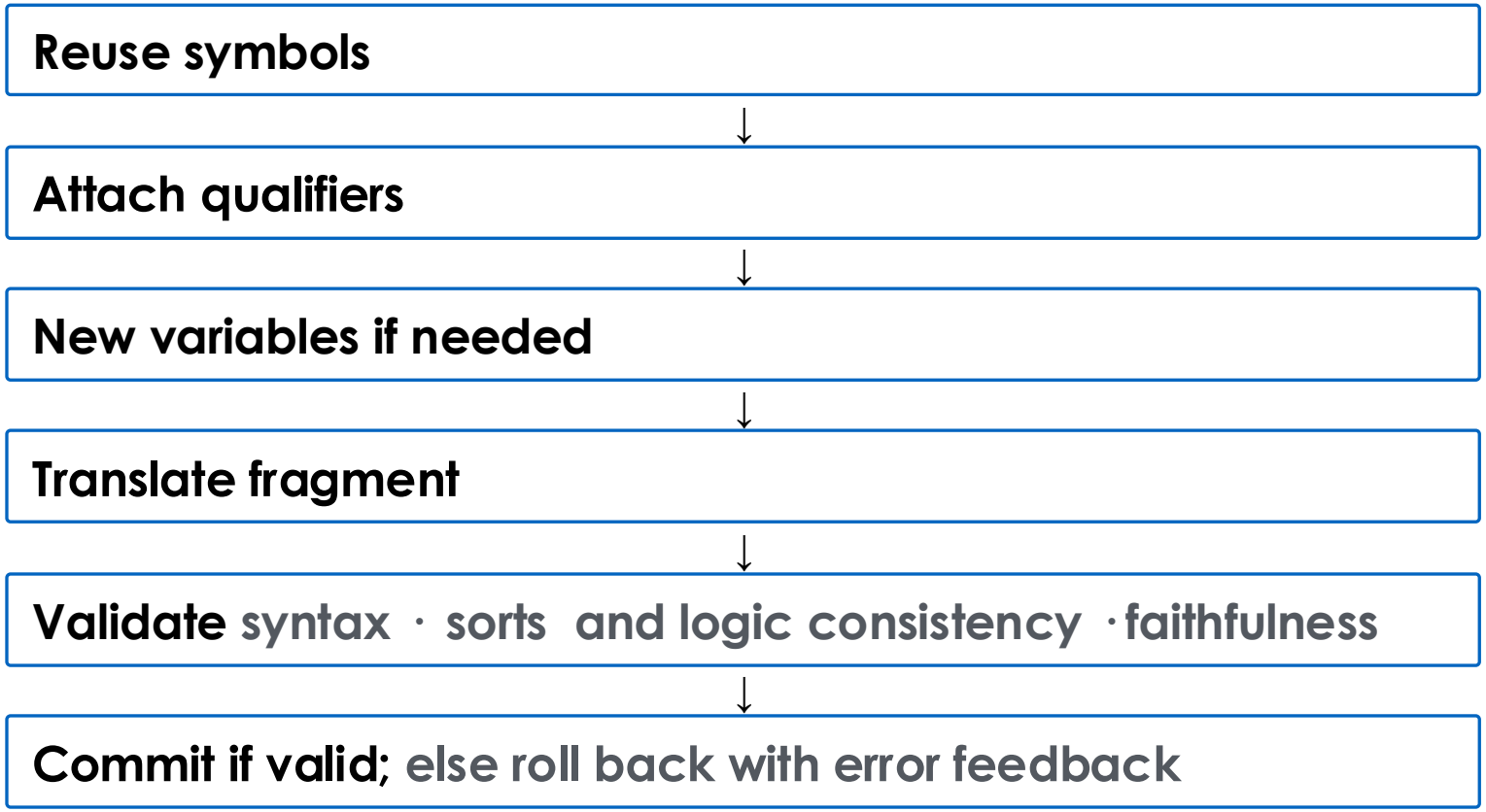
49 **patients** **2,578** **source fact lines** → **48,845** **SMT fact lines**

patient facts after ontology enrichment (TREC, 49 patients)

Implementation Complexities

Each requirement ↔ its SMT lines: errors are attributable to a sentence.

Many steps to ensure the SMT program is correct



Summary: Disadvantages of SMT Approach

- SMT solvers operate on variables – they have no meaning
- All information useful for SMT solvers must be included formally

1. **Taxonomy**: same concepts → same variable name **Canonicalization**

2. **Implication relations** between concepts
   e.g. A ⇒ B means A satisfies B
   Include implication relations

3. **Missing Data:** most conditions in trials are missing **Data Imputation**

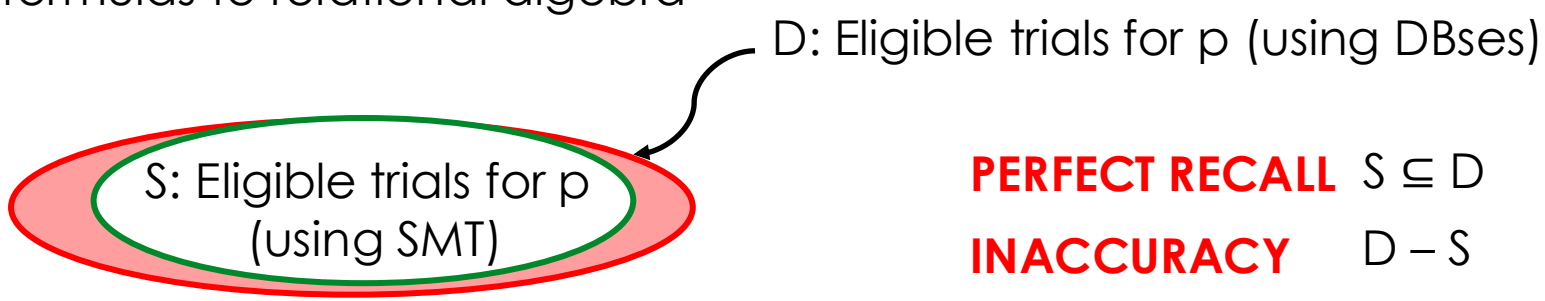
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Projection of SMT to Relational Algebra

SatIR: Use relational algebra (Databases): **less expressive**, but **much cheaper**

Project SMT formulas to relational algebra



PERFECT RECALL $S \subseteq D$

INACCURACY $D - S$

SMT vs. Propositional Logic (PL) example

At least three of the following disorders: A, B, C, D, E, F, G

SMT

```
(assert (>= (+
            (ite A 1 0)
            (ite B 1 0)
            (ite C 1 0)
            (ite D 1 0)
            (ite E 1 0)
            (ite F 1 0)
            (ite G 1 0))
        3))
```

Ite: If This Then

PL

(A∧B∧C) ∨ (A∧B∧D) ∨ (A∧B∧E) ∨ (A∧B∧F)
∨ (A∧B∧G) ∨(A∧C∧D) ∨ (A∧C∧E) ∨ (A∧C∧F)
∨ (A∧C∧G) ∨(A∧D∧E) ∨ (A∧D∧F) ∨ (A∧D∧G)
∨ (A∧E∧F) ∨(A∧E∧G) ∨ (A∧F∧G) ∨ (B∧C∧D)
∨ (B∧C∧E) ∨(B∧C∧F) ∨ (B∧C∧G) ∨ (B∧D∧E)
∨ (B∧D∧F) ∨(B∧D∧G) ∨ (B∧E∧F) ∨ (B∧E∧G)
∨ (B∧F∧G) ∨(C∧D∧E) ∨ (C∧D∧F) ∨ (C∧D∧G)
∨ (C∧E∧F) ∨(C∧E∧G) ∨ (C∧F∧G) ∨ (D∧E∧F)
∨ (D∧E∧G) ∨(D∧F∧G) ∨ (E∧F∧G)

SMT is direct and succinct

Some requirements representable in SMT cannot be expressed in propositional logic
(e.g. counting incidents in a time window)

Projection designed to not lose recall

sacrificing precision if necessary

```
(and T2DM@@within5y
     (>= hba1c 7.0) (<= hba1c 10.0)
     HasAdequateSocialSupportForFollowUp)
```

```
(and T2DM@@within5y
     (>= hba1c 7.0) (<= hba1c 10.0))
; non-canonical atom projected away
```

A patient without social support matches
→ no loss in recall, loss in precision

Quiz: Why do we bother to include non-canonical atoms?

- In retrieval:

```
(OR  T2DM@@within5y
     (>= hba1c 7.0) (<= hba1c 10.0)
     HasAdequateSocialSupportForFollowUp)
```

- In matching:

Projecting SMT Matching to Database Queries

1. **Project away** non-canonical parts

2. **Represent as Canonical Normal Form** (AND of ORs of atoms)

3. **Match = Table** joins + aggregation

Projection: every step is conservative (in the paper)

Theorem:
SatIR has perfect recall (assuming the SMT formulation is correct)

Representing PL formulas in DB: Example

c1 {
(***has_bipolar*** OR ***has_depression*** OR ***has_schizo***)
AND (NOT ***has_undergone_PCI***)
AND ((NOT ***is_smoking***) OR (NOT ***is_drinking***))

a1
a2
a3

C	∧
c1	a1
c1	a2
c1	a3
c2	…
c2	…
c3	…

∧	∨
a1	o1
a1	o2
a1	o3
a2	o4
a3	o5
a3	o6

∨	Variables	Is Pos
o1	has_bipolar	1
o2	has_depression	1
o3	has_schizo	1
o4	has_undergone_PCI	0
o5	is_smoking	0
o6	is_drinking	0

Constraints

∧ (and) clauses

∨ (or) clauses

Matching Patient 1

has_bipolar AND ***has_depression***

(***has_bipolar*** OR ***has_depression*** OR ***has_schizo***)
AND (NOT ***has_undergone_PCI***)
AND ((NOT ***is_smoking***) OR (NOT ***is_drinking***))

pid	Variable	Val
123	has_bipolar	1
123	has_depression	1
123	has_schizo	0
123	has_undergone_PCI	0
123	is_smoking	0
123	Is_drinking	0

Patient Fact Table

∨	Variable	Is Pos
o1	has_bipolar	1
o2	has_depression	1
o3	has_schizo	1
o4	has_undergone_PCI	0
o5	is_smoking	0
o6	is_drinking	0

∨ (or) clauses

∧	∨
a1	o1
a1	o2
a1	o3
a2	o4
a3	o5
a3	o6

∧ (and) clauses

C	∧
c1	a1
c1	a2
c1	a3

Constraints



+ Eval literals
(SQL query)

Matching Patient 1

has_bipolar AND ***has_depression***

(***has_bipolar*** OR ***has_depression*** OR ***has_schizo***)
AND (NOT ***has_undergone_PCI***)
AND ((NOT ***is_smoking***) OR (NOT ***is_drinking***))

pid	v	Variable	Val	Is_pos	Eval
123	o1	has_bipolar	1	1	1
123	o2	has_depression	1	1	1
123	o3	has_schizo	0	1	0
123	o4	has_undergone_PCI	0	0	1
123	o5	is_smoking	0	0	1
123	o6	is_drinking	0	0	1

literal-Evaluated



+ Eval v
(SQL query)

∧	v
a1	o1
a1	o2
a1	o3
a2	o4
a3	o5
a3	o6

∧ (and) clauses

C	∧
c1	a1
c1	a2
c1	a3

Constraints

Matching Patient 1

has_bipolar AND ***has_depression***

(***has_bipolar*** OR ***has_depression*** OR ***has_schizo***)
AND (NOT ***has_undergone_PCI***)
AND ((NOT ***is_smoking***) OR (NOT ***is_drinking***))

∧	eval
a1	1
a2	1
a3	1

v evaluated



Eval ∧
(SQL query)

C	∧
c1	a1
c1	a2
c1	a3

Constraints

C	C_eval
C1	**1**

SAT!

Matching Patient (unsat)

has_schizo AND ***is_smoking***
AND ***is_drinking***

(***has_bipolar*** OR ***has_depression*** OR ***has_schizo***)
AND (NOT ***has_undergone_PCI***)
AND ((NOT ***is_smoking***) OR (NOT ***is_drinking***))

∧	∧_eval
a1	1
a2	1
a3	0

C	∧
c1	a1
c1	a2
c1	a3

C	C_eval
C1	**0**

UNSAT!

∧ (and) Evaluated



Constraints

+Eval ∧
(SQL query)

Theorem: projection never loses recall

The price: precision

Relational constraints are less expressive to evaluate; a downstream matcher re-checks candidates.

146 ms

per patient · 3,621 trials · laptop

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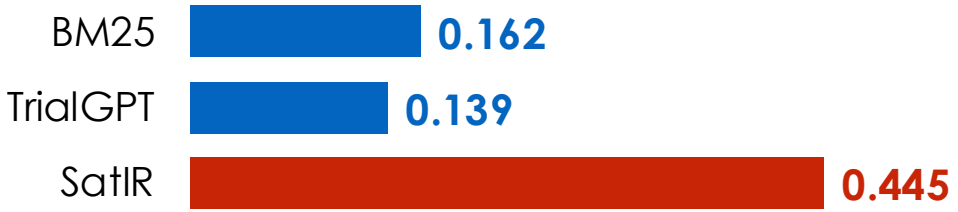
More of the right trials

Labels: GPT-5 judge; validated a sample by 2 clinicians, strong agreement

SIGIR 2016, treat-chief (59 patients, 3621 trials)
Relevant & eligible per patient



TREC 2022 subset (50 patients, 2658 trials)
recall, treat-chief, qrel = 2 (highly relevant)



Union of good trials across techniques



Where SatIR still fails

74

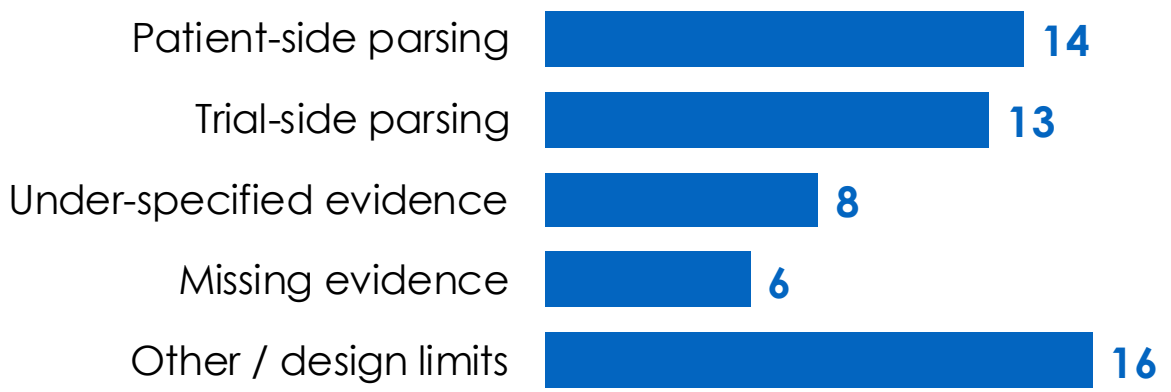
found only by TrialGPT
(200 candidates per patient)

Error Analysis →

253

found only by SatIR

57 clear failures, by source




```
"age over 15"  →  (> age 15)    ; patient is 15
should it be (>= age 15)?  -- visible, arguable, fixable
```

SatIR: a new retriever based on constraint satisfaction

Constraints in documents are stored as first-class objects in databases

1. **Be conservative**: don't let unformalizable content exclude candidates
2. Compile **domain knowledge** into constraints (largest effect)
3. Store **missing-data assumptions** as data you can inspect and ablate
4. Build programs **incrementally**, **validated** at every step
5. Relax technique for speed if necessary

   -- only with a **proof** of the properties you maintain